

Research Question Sharpener

Fill this out before writing a single line of code. If you can't complete it, the project isn't ready.

1 THE WHAT

What are you claiming or showing?

Be specific enough that a smart person could push back. Not "we investigate X" but "we show that X, which matters because Y."

The Surprise

What prior belief or expectation does this violate? Who specifically would update their thinking based on this result? If the answer is "nobody" — the question isn't interesting enough yet.

One-Sentence Version

Force yourself to state the contribution in a single sentence. If you can't do this clearly, the idea isn't sharp enough yet.

2 THE WHY — Fruitfulness

A fruitful result doesn't just answer one question — it opens a research program. Fruitfulness is what separates a dead-end finding from a discovery that matters.

High fruitfulness (crossroads)	Low fruitfulness (dead end)
Grokking: Nets suddenly generalize long after memorizing. Opened: why the delay? Phase transition? Does it happen in large models? Connection to double descent? Adversarial examples: One finding spawned an entire subfield — robustness, certified defenses, loss landscapes, alignment implications. Sycophancy paper: Users prefer the thing that harms them → opens: long-term effects? Other domains? Cognitive mechanism? Policy?	"We beat SOTA by 2%": Correct but nobody builds on it. No mechanism to explore, no follow-up questions emerge. "LLMs are worse in Swahili": Confirms expectations from training data. Next step (fix it) is engineering, not science. "Model X does task Y": Unless this was thought impossible or reveals how X works, it's a capability report, not a discovery.

New questions this result would open

List 2-4 follow-up questions that become askable because of your result. Then ask yourself honestly: are these interesting BECAUSE of your finding, or were they already obvious questions anyone could ask without your paper?

Name specific researchers, labs, or communities who would use your result as a stepping stone for their own work. If you can't name anyone, the result may be a dead end.

Does your result open multiple directions of inquiry, or just one? Does it connect to other fields or stay within its own lane? The best results are crossroads — they give different communities different things to chase.

Design experiments that make your result undeniable. Every experiment should be traceable to a specific alternative explanation it forecloses.

What's the simplest way a skeptic dismisses this? "It's just training data / scale / cherry-picking / a confound." What control or ablation forecloses that escape route?

What experiments do you run? What are the manipulations, controls, measurements, and success criteria? How do you show this is robust and not a fluke?

Seeds/trials, statistical tests, preregistration, ablations. What would leave a reviewer with nothing to grab onto?

Depth Level of Your Design

What is your experimental design CAPABLE of producing? If it can only show "X happens," no result will magically give you Level 3.

Level	Your design can show...	✓
0	We propose X, it works	
1	X works / exists + strong evidence it's real	
2	WHY X works + reveals structure of the problem	
3	General principle that reframes broader thinking	

If Level 0-1: can you redesign to reach Level 2? Aim for 2+.

Impact Type

■ Conceptual (changes how experts think) · ■ Practical (enables something previously impossible) · ■ Methodological (tools others will adopt)

Broadest Truthful Audience

Who outside your immediate field would care? What framing reaches them without overstating?

The 'Science Version'

If this were in Science or Nature, what would the title be? What would the press release lead with?